

Polar to Cartesian: $x = r \cos \theta$ $y = r \sin \theta$

Cartesian to Polar: $r = \sqrt{x^2 + y^2}$ $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

Volume of a function, $f(r, \theta)$ is given by:

$$V = \int_{\theta_1}^{\theta_2} \int_{r_1(\theta)}^{r_2(\theta)} f(r, \theta) \cdot r \, dr \, d\theta$$

★ this is new

↳ This is used when R is a circular region or the sector of a circle or when the equation/limits are simpler in polar form.

(e.g.) Use polar coordinates to evaluate the double integral: $\int_{-a}^a \int_0^{\sqrt{a^2 - x^2}} (x^2 + y^2)^{1/2} \, dy \, dx$

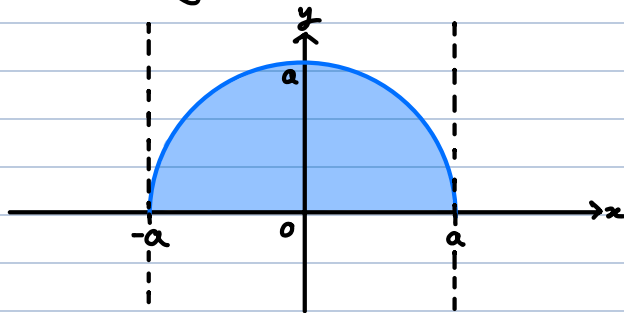
Solⁿ: From the limits we first need to identify the region R .

→ Since the outer limits are limits of x , this is a type I region (the region is bound on the left and right sides by the vertical lines $x = -a$ and $x = a$).

→ The lower limit of y is 0 and the upper limit is $\sqrt{a^2 - x^2}$. So the region is bounded below by the line $x = 0$ and above by the curve:

$$y = \sqrt{a^2 - x^2} \Rightarrow y^2 + x^2 = a^2 \quad \leftarrow \text{Equation of a circle of radius "a"}$$

→ Hence, the region R is:



← A semi-circle
of radius "a"

→ So converting the double integral to polar form:

The function $(x^2 + y^2)^{1/2}$

becomes $(r^2)^{1/2} = r$

← $r^2 = x^2 + y^2$

The angle (θ) spans from 0 radians to π radians.

↖ 180°

The radius (r) goes from 0 to "a",

since the outer boundary is a circle with

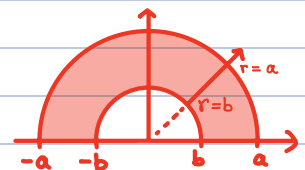
the equation $r = a$, and there is no inner

circle (i.e. the inner circle has the equation $r = 0$).

★ If the region was something like:

then r would have increased

radially from $r = b$ to $r = a$.



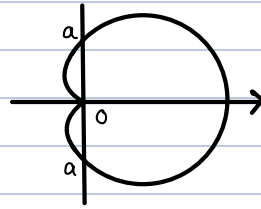
Hence, the new integral is:

$$\int_0^\pi \int_0^a (r) r dr d\theta = \int_0^\pi \int_0^a r^2 dr d\theta = \int_0^\pi \left[\frac{r^3}{3} \right]_0^a d\theta$$

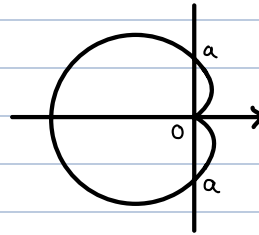
$$= \int_0^\pi \frac{a^3}{3} d\theta = \left[\frac{a^3 \theta}{3} \right]_0^\pi = \frac{a^3 \pi}{3}$$

Some common polar curves (Cartoids):

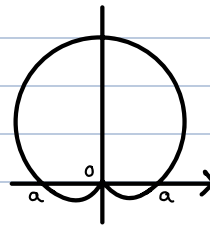
$$\rightarrow r = a(1 + \cos \theta)$$



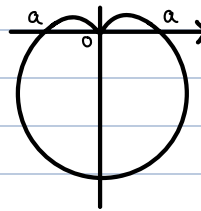
$$\rightarrow r = a(1 - \cos \theta)$$



$$\rightarrow r = a(1 + \sin \theta)$$

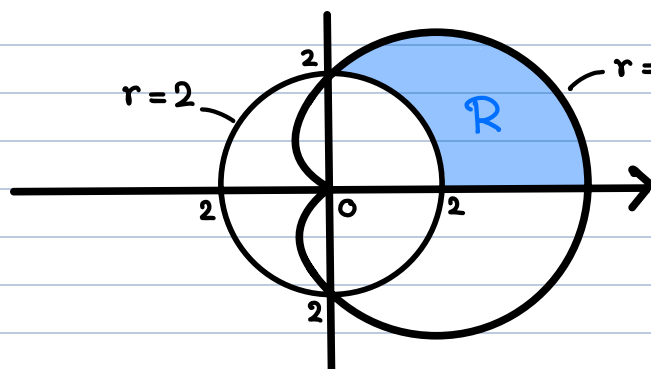


$$\rightarrow r = a(1 - \sin \theta)$$



(e.g.) Evaluate $\iint_R \sin \theta \, dA$ where R is the region in the first quadrant that's outside the circle $r=2$ and inside the cardioid $r=2(1+\cos \theta)$.

Solⁿ: The region R is the shaded region:



$r=2(1+\cos \theta)$ ← The region R is:

↳ in the 1st quadrant

↳ outside circle $r=2$

↳ inside cardioid
 $r=2(1+\cos \theta)$

So integrating over this region, we have:

$$\begin{aligned}& \int_0^{\frac{\pi}{2}} \int_2^{2(1+\cos\theta)} \sin\theta \, r \, dr \, d\theta \\&= \int_0^{\frac{\pi}{2}} \left[\frac{r^2}{2} \cdot \sin\theta \right]_2^{2(1+\cos\theta)} d\theta = \int_0^{\frac{\pi}{2}} 2(1+\cos\theta)^2 \cdot \sin\theta - 2\sin\theta \, d\theta \\&= \int_0^{\frac{\pi}{2}} 2(1+2\cos\theta+\cos^2\theta)\sin\theta - 2\sin\theta \, d\theta \\&= \int_0^{\frac{\pi}{2}} \cancel{2\sin\theta} + 4\sin\theta\cos\theta + 2\sin\theta\cos^2\theta - \cancel{2\sin\theta} \, d\theta \\&= \int_0^{\frac{\pi}{2}} 2\sin 2\theta + 2\sin\theta(1-\sin^2\theta) \, d\theta \\&= \int_0^{\frac{\pi}{2}} 2\sin 2\theta + 2\sin\theta - 2\sin^3\theta \, d\theta \\&= \left[-\cos 2\theta - 2\cos\theta \right]_0^{\frac{\pi}{2}} - 2 \int_0^{\frac{\pi}{2}} \sin^3\theta \, d\theta \\&= 1 - (-1-2) - 2 \int_0^{\frac{\pi}{2}} (1-\cos^2\theta)\sin\theta \, d\theta \\&= 4 - 2 \int_1^0 (1-u^2) \cancel{\sin\theta} \cdot \frac{du}{\cancel{-\sin\theta}} \\&= 4 - 2 \int_0^1 1-u^2 \, du = 4 - 2 \left[u - \frac{u^3}{3} \right]_0^1 \\&= 4 - 2 \left(1 - \frac{1}{3} \right) = 4 - \frac{4}{3} = \frac{8}{3}\end{aligned}$$

Let $u = \cos\theta$

$$\frac{du}{d\theta} = -\sin\theta$$

$$\Rightarrow d\theta = \frac{du}{-\sin\theta}$$

At $\theta=0$, $u=1$

At $\theta=\frac{\pi}{2}$, $u=0$

Triple integrals: $\iiint_V f(x,y,z) dx dy dz$ $\leftarrow$ integration over a volume

Like double integrals, for triple integrals:

↳ Volume of integration V can be divided into $V = V_1 + V_2$

↳ Can be non-uniform (not a box).

↳ Can give volume by setting the integrand to 1.

(e.g.) Find the volume of the solid that's bounded by the cylinder $y = x^2$ and by the planes $y + z = 4$ and $z = 0$.

Solⁿ: $\rightarrow$ Since we need to calculate the volume, take the integrand to be 1.

$\rightarrow$ The innermost integral should be with respect to z .

$\rightarrow$ As the volume is bounded by the planes $z = 4 - y$ and $z = 0$, the limits of z are $\int_0^{4-y}$

$\therefore$ The innermost integral is:

$$\int_0^{4-y} 1 dz.$$

$\rightarrow$ The integral outside this will be of the form:

$$\iint_R dA$$

which is a double integral where the region R is the intersection of the solid with the xy -plane.

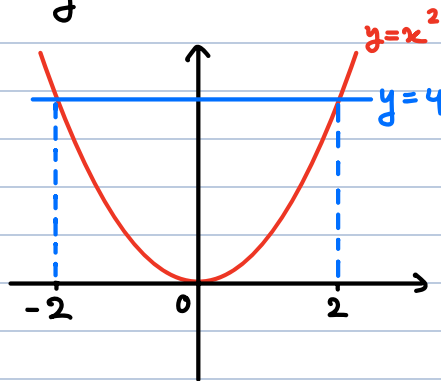
→ At the xy -plane, $z=0$. The boundaries of the solid are $y=x^2$, $y+z=4$.

→ From these equations, for each one we set $z=0$.

$\therefore y+z=4$, set $z=0$ ← since $z=0$ in the xy -plane.

$\Rightarrow y=4$. Then we draw $y=x^2$ and $y=4$ in the xy -plane. The region bounded by these is the region R .

→ The plane $y+z=4$ intersects the plane $z=0$ at the line $y=4$.



← At $y=4$,
 $x^2=4 \Rightarrow x=\pm 2$

→ So the limits of y are $\int_{x^2}^4$ and the limits of x are $\int_{-2}^2$. Hence, the integral is:

$$\begin{aligned} V &= \int_{-2}^2 \int_{x^2}^4 \int_0^{4-y} 1 \, dz \, dy \, dx \\ &= \int_{-2}^2 \int_{x^2}^4 [z]_0^{4-y} \, dy \, dx = \int_{-2}^2 \int_{x^2}^4 (4-y) \, dy \, dx \\ &= \int_{-2}^2 \left[4y - \frac{y^2}{2} \right]_{x^2}^4 \, dx = \int_{-2}^2 \left(16 - \frac{16}{2} \right) - \left(4x^2 - \frac{x^4}{2} \right) \, dx \\ &= \int_{-2}^2 \left(8 - 4x^2 + \frac{x^4}{2} \right) \, dx = \left[8x - \frac{4x^3}{2} + \frac{x^5}{10} \right]_{-2}^2 \end{aligned}$$

$$= \left(16 - \frac{4 \times 8}{2} + \frac{32}{10} \right) - \left(-16 + \frac{4 \times 8}{2} - \frac{32}{10} \right)$$

$$= \frac{16}{5} + \frac{16}{5} = \frac{32}{5} //$$